Topic: Simple Harmonic Motion, Energy in SHM, Damped Simple Harmonic Motion

What is Simple Harmonic Motion?

Simple Harmonic Motion (SHM) is a type of periodic motion where the restoring force is directly proportional to the displacement from the equilibrium position and acts in the opposite direction.

Characteristics of SHM

- The restoring force follows Hooke's Law:

$$F = -kx \tag{1}$$

  where k is the force constant and x is the displacement.

- The displacement as a function of time is:

$$x(t) = x_m \cos(\omega t + \phi) \tag{2}$$

  where:

  - x_m is the amplitude,
  - ω is the angular frequency, and
  - ϕ is the phase constant.

- The velocity of the oscillating particle is derived by differentiating displacement:

$$v(t) = \frac{dx}{dt} = \frac{d}{dt}\left(x_m \cos(\omega t + \phi)\right) \tag{3}$$

  Using the derivative of cosine:

$$v(t) = -x_m \omega \sin(\omega t + \phi) \tag{4}$$

- The acceleration is derived by differentiating velocity:

$$a(t) = \frac{dv}{dt} = \frac{d}{dt}\left(-x_m \omega \sin(\omega t + \phi)\right) \tag{5}$$

Using the derivative of sine:

$$a(t) = -x_m\omega^2 \cos(\omega t + \phi) \tag{6}$$

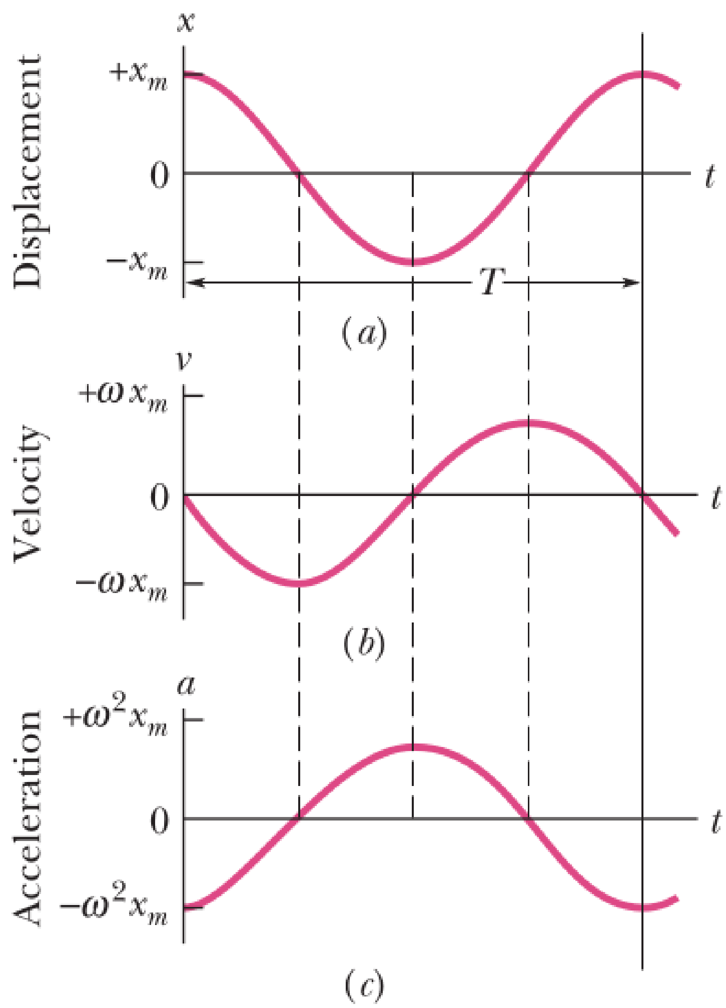


Figure 1: (a) The displacement x(t) of a particle oscillating in SHM with phase angle zero. The period T marks one complete oscillation. (b) The velocity v(t) of the particle. (c) The acceleration a(t) of the particle.

- The time period T of SHM is the time taken for one complete oscillation and is given by:

$$T = \frac{2\pi}{\omega} \tag{7}$$

  Since the angular frequency is:

$$\omega = \sqrt{\frac{k}{m}} \tag{8}$$

  The time period can be written as:

$$T = 2\pi\sqrt{\frac{m}{k}} \tag{9}$$

- The frequency f of SHM is the number of oscillations per unit time and is given by:

$$f = \frac{1}{T} \tag{10}$$

  Substituting T:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \tag{11}$$

Differential Equation of SHM

The force acting on a particle undergoing SHM is given by Hooke's Law:

$$F = -kx \tag{12}$$

Using Newton's second law, we have:

$$ma = -kx \tag{13}$$

Since acceleration is the second derivative of displacement with respect to time:

$$m\frac{d^2x}{dt^2} = -kx \tag{14}$$

Rearranging, we obtain the standard form of the differential equation of SHM:

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \tag{15}$$

Defining the angular frequency as:

$$\omega = \sqrt{\frac{k}{m}} \tag{16}$$

The differential equation simplifies to:

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \tag{17}$$

This is the differential equation governing Simple Harmonic Motion.

Energy in SHM

The total mechanical energy of a simple harmonic oscillator is given by the sum of its kinetic and potential energies.

Potential Energy

The potential energy stored in the spring is given by:

$$U(t) = \frac{1}{2}kx^2 \tag{18}$$

Substituting $x(t) = x_m \cos(\omega t + \phi)$:

$$U(t) = \frac{1}{2}kx_m^2 \cos^2(\omega t + \phi) \tag{19}$$

Kinetic Energy

The kinetic energy of the mass is given by:

$$K(t) = \frac{1}{2}mv^2 \tag{20}$$

Substituting $v(t) = -\omega x_m \sin(\omega t + \phi)$:

$$K(t) = \frac{1}{2}m\omega^2 x_m^2 \sin^2(\omega t + \phi) \tag{21}$$

Using $\omega^2 = \frac{k}{m}$:

$$K(t) = \frac{1}{2}kx_m^2 \sin^2(\omega t + \phi) \tag{22}$$

Total Mechanical Energy

The total energy of the system is:

$$E = U + K \tag{23}$$

Substituting the expressions for $U(t)$ and $K(t)$:

$$E = \frac{1}{2}kx_m^2 \cos^2(\omega t + \phi) + \frac{1}{2}kx_m^2 \sin^2(\omega t + \phi) \tag{24}$$

Using the identity $\cos^2 a + \sin^2 a = 1$:

$$E = \frac{1}{2}kx_m^2 \tag{25}$$

Thus, the total mechanical energy remains constant throughout the motion.

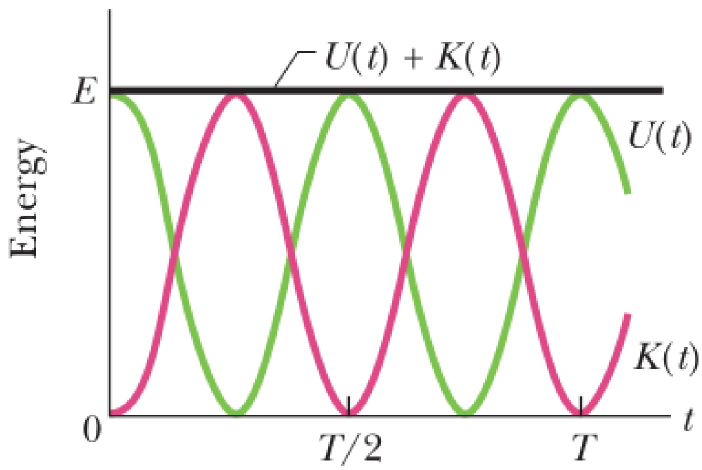


Figure 2: Potential energy U(t), kinet- ic energy K(t), and mechanical energy E as functions of time t for a linear harmon- ic oscillator

Problem 1:

A block of mass $m = 680$ g is attached to a spring with a spring constant $k = 65$ N/m. The block is pulled a distance $x_m = 11$ cm from its equilibrium position and released from rest at $t = 0$. The system undergoes simple harmonic motion (SHM) on a frictionless surface.

(a) **Find the angular frequency ω, the frequency f, and the period T of the motion.**

(b) **Determine the amplitude of oscillation.**

(c) **Find the maximum speed v_m of the block and the position where this occurs.**

(d) **Determine the maximum acceleration a_m of the block.**

(e) **Find the phase constant ϕ for the motion.**

(f) **Write the displacement function $x(t)$ for the motion.**

Solution

(a) Angular Frequency, Frequency, and Period

The angular frequency is given by:

$$\omega = \sqrt{\frac{k}{m}} \tag{26}$$

Substituting $k = 65$ N/m and $m = 0.68$ kg:

$$\omega = \sqrt{\frac{65}{0.68}} = 9.78 \text{ rad/s} \tag{27}$$

The frequency is given by:

$$f = \frac{\omega}{2\pi} \tag{28}$$

$$f = \frac{9.78}{2\pi} = 1.56 \text{ Hz} \approx 1.6 \text{ Hz} \tag{29}$$

The period is:

$$T = \frac{1}{f} \tag{30}$$

$$T = \frac{1}{1.56} = 0.64 \text{ s} = 640 \text{ ms} \tag{31}$$

(b) Amplitude of Oscillation

The amplitude of oscillation is simply the maximum displacement:

$$x_m = 11 \text{ cm} = 0.11 \text{ m} \tag{32}$$

(c) Maximum Speed

The maximum speed occurs at equilibrium ($x = 0$) and is given by:

$$v_m = \omega x_m \tag{33}$$

$$v_m = (9.78)(0.11) = 1.1 \text{ m/s} \tag{34}$$

(d) Maximum Acceleration

The maximum acceleration is given by:

$$a_m = \omega^2 x_m \tag{35}$$

$$a_m = (9.78)^2(0.11) = 11 \text{ m/s}^2 \tag{36}$$

(e) Phase Constant

Since the block is released from rest at $x = x_m$, we set the general displacement function:

$$x(t) = x_m \cos(\omega t + \phi) \tag{37}$$

At $t = 0$, we have $x(0) = x_m$, so:

$$x_m = x_m \cos \phi \tag{38}$$

Dividing by x_m:

$$1 = \cos \phi \tag{39}$$

Thus,

$$\phi = 0 \text{ rad} \tag{40}$$

(f) Displacement Function

Using the values obtained:

$$x(t) = (0.11) \cos(9.8t) \tag{41}$$

where x is in meters and t is in seconds.

Problem 2:

A block of mass $m = 2.72 \times 10^5$ kg is attached to a spring and oscillates with a frequency of $f = 10.0$ Hz and an amplitude of $x_m = 20.0$ cm.

(a) Find the total mechanical energy E of the system.

(b) Determine the block's speed as it passes through the equilibrium position.

Solution

(a) Total Mechanical Energy

The total mechanical energy E in simple harmonic motion is given by:

$$E = \frac{1}{2} k x_m^2 \tag{42}$$

where k is the spring constant. To find k, we use the relation:

$$\omega^2 = \frac{k}{m} \tag{43}$$

Since $\omega = 2\pi f$, we get:

$$k = m(2\pi f)^2 \tag{44}$$

Substituting the given values:

$$k = (2.72 \times 10^5)(2\pi \times 10.0)^2 \tag{45}$$

$$k = 1.073 \times 10^9 \text{ N/m} \tag{46}$$

Now, substituting k and $x_m = 0.20$ m into the energy equation:

$$E = \frac{1}{2}(1.073 \times 10^9)(0.20)^2 \tag{47}$$

$$E = 2.147 \times 10^7 \text{ J} \approx 2.1 \times 10^7 \text{ J} \tag{48}$$

(b) Speed at Equilibrium Position

At equilibrium ($x = 0$), all the energy is kinetic:

$$E = \frac{1}{2}mv^2 \tag{49}$$

Solving for v:

$$v = \sqrt{\frac{2E}{m}} \tag{50}$$

Substituting the values:

$$v = \sqrt{\frac{2(2.147 \times 10^7)}{2.72 \times 10^5}} \tag{51}$$

$$v = 12.6 \text{ m/s} \tag{52}$$

Thus, the maximum speed of the block is 12.6 m/s.

Problem 3:

At $t = 0$, the displacement $x(0)$ of the block in a linear oscillator is $8.50\,\mathbf{cm}$. The block's velocity $v(0)$ then is $0.920\,\mathbf{m/s}$, and its acceleration $a(0)$ is $-47.0\,\mathbf{m/s^2}$. Determine the angular frequency ω, phase constant φ, and amplitude x_m.

Solution

The displacement, velocity, and acceleration of the block in Simple Harmonic Motion (SHM) are given by the following equations:

$$x(t) = x_m \cos(\omega t + \varphi),$$
$$v(t) = -\omega x_m \sin(\omega t + \varphi),$$
$$a(t) = -\omega^2 x_m \cos(\omega t + \varphi).$$

At $t = 0$, we have:

$$x(0) = x_m \cos(\varphi), \tag{1}$$

$$v(0) = -\omega x_m \sin(\varphi), \tag{2}$$

$$a(0) = -\omega^2 x_m \cos(\varphi). \tag{3}$$

From the problem statement, we know the following values at $t = 0$:

$$x(0) = 8.50\,\text{cm} = 0.0850\,\text{m},$$

$$v(0) = 0.920\,\text{m/s},$$

$$a(0) = -47.0\,\text{m/s}^2.$$

(a) Determining the angular frequency ω:

To solve for ω, we divide equation (3) by equation (1):

$$\frac{a(0)}{x(0)} = \frac{-\omega^2 x_m \cos(\varphi)}{x_m \cos(\varphi)} = -\omega^2,$$

which simplifies to:

$$\omega^2 = \frac{-a(0)}{x(0)}.$$

Substituting the known values:

$$\omega^2 = \frac{-(-47.0)}{0.0850} = \frac{47.0}{0.0850} = 552.9.$$

Thus:

$$\omega = \sqrt{552.9} = 23.5\,\text{rad/s}.$$

(b) Determining the phase constant φ and amplitude x_m:

From Equation 1 and 2, we have:

$$x(0) = x_m \cos(\varphi),$$

$$v(0) = -\omega x_m \sin(\varphi).$$

Dividing Equation (2) by Equation (1):

$$\frac{v(0)}{x(0)} = \frac{-\omega x_m \sin(\varphi)}{x_m \cos(\varphi)} = -\omega \tan(\varphi).$$

Substituting the known values for $v(0) = 0.920\,\text{m/s}$ and $x(0) = 0.0850\,\text{m}$:

$$\frac{0.920}{0.0850} = -23.5 \tan(\varphi),$$

$$10.82 = -23.5 \tan(\varphi),$$

$$\tan(\varphi) = -\frac{10.82}{23.5} = -0.461.$$

Taking the inverse tangent:

$$\varphi = \tan^{-1}(-0.461).$$

This gives solution for the phase constant:

$$\varphi = -25^\circ$$

Next, we use the displacement equation to determine x_m. From Equation (1):

$$x(0) = x_m \cos(\varphi),$$

$$x_m = \frac{x(0)}{\cos(\varphi)}.$$

For value of φ, we find:

$$x_m = \frac{0.0850}{\cos(-25^\circ)} = 0.094\,\text{m}.$$

Therefore, the amplitude is $x_m = 0.094\,\text{m}$, or $9.4\,\text{cm}$.

Spring Constant in Series and Parallel

Springs in Series

When springs are connected in series, the total spring constant k_{total} is given by the reciprocal of the sum of the reciprocals of the individual spring constants. Mathematically, we can write this as:

$$\frac{1}{k_{\text{total}}} = \frac{1}{k_1} + \frac{1}{k_2} + \cdots + \frac{1}{k_n}$$

Or equivalently:

$$k_{\text{total}} = \left(\frac{1}{k_1} + \frac{1}{k_2} + \cdots + \frac{1}{k_n} \right)^{-1}$$

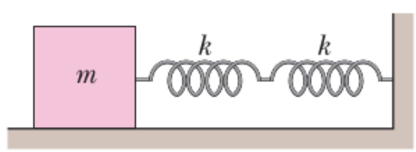
Springs in Parallel

When springs are connected in parallel, the total spring constant k_{total} is simply the sum of the individual spring constants. This is given by:

$$k_{\text{total}} = k_1 + k_2 + \cdots + k_n$$

Problem 4:

Two springs, each with spring constant $k = 6430\,\mathrm{N/m}$, are joined and connected to a block of mass $m = 0.245\,\mathrm{kg}$ that is set oscillating over a frictionless floor. The springs are connected in series. What is the frequency of the oscillations?



Solution

1. Springs in Series:

When two springs are connected in series, the total spring constant k_{total} is given by the reciprocal of the sum of the reciprocals of the individual spring constants:

$$\frac{1}{k_{\text{total}}} = \frac{1}{k_1} + \frac{1}{k_2}$$

Since both springs have the same spring constant $k = 6430\,\mathrm{N/m}$, we have:

$$\frac{1}{k_{\text{total}}} = \frac{1}{6430} + \frac{1}{6430} = \frac{2}{6430}$$

Taking the reciprocal:

$$k_{\text{total}} = \frac{6430}{2} = 3215\,\text{N/m}$$

2. Frequency of Oscillation:

The frequency of oscillation for a mass-spring system is given by the formula:

$$f = \frac{1}{2\pi}\sqrt{\frac{k_{\text{total}}}{m}}$$

Substituting the known values of k_{total} and m:

$$f = \frac{1}{2\pi}\sqrt{\frac{3215}{0.245}} = \frac{1}{2\pi}\sqrt{13102.04}$$

Now, calculate the square root:

$$\sqrt{13102.04} \approx 114.5$$

Thus:

$$f = \frac{1}{2\pi} \times 114.5$$

Finally, divide by 2π:

$$f \approx \frac{114.5}{6.2832} \approx 18.2\,\text{Hz}$$

Thus, the frequency of the oscillations is approximately:

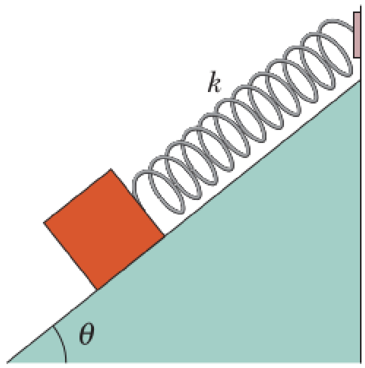
$$f \approx 18.2\,\text{Hz}$$

Problem 5:

A block weighing $14.0\,\mathrm{N}$, which can slide without friction on an incline at an angle $\theta = 40.0°$, is connected to the top of the incline by a massless spring of unstretched length $0.450\,\mathrm{m}$ and spring constant $120\,\mathrm{N/m}$.

(a) How far from the top of the incline is the block's equilibrium point?

(b) If the block is pulled slightly down the incline and released, what is the period of the resulting oscillations?



Solution

(a) Equilibrium Position

To determine the equilibrium position, we consider the forces acting on the block along the incline. When the block is at equilibrium, the restoring force of the spring balances the component of the gravitational force along the incline:

$$kx = mg\sin\theta$$

Solving for x:

$$x = \frac{mg\sin\theta}{k}$$

Given:

$$mg = 14.0\,\mathrm{N}, \quad k = 120\,\mathrm{N/m}, \quad \theta = 40.0°$$

Substituting the values:

$$x = \frac{(14.0\,\mathrm{N})\sin 40.0°}{120\,\mathrm{N/m}}$$

$$x = \frac{14.0 \times 0.6428}{120}$$

$$x = \frac{8.9992}{120} = 0.0750\,\mathrm{m}$$

Thus, the equilibrium position of the block is at a distance:

$$d = 0.450 + 0.0750 = 0.525\,\mathrm{m}$$

from the top of the incline.

(b) Period of Oscillation

The period of oscillation for a mass-spring system is given by:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Since the weight of the block is given instead of mass, we use:

$$m = \frac{W}{g} = \frac{14.0\,\mathrm{N}}{9.80\,\mathrm{m/s}^2} = 1.4286\,\mathrm{kg}$$

Now, substituting the values:

$$T = 2\pi\sqrt{\frac{1.4286}{120}}$$

$$T = 2\pi\sqrt{0.0119}$$

$$T = 2\pi \times 0.1090$$

$$T \approx 0.686\,\mathrm{s}$$

Thus, the period of oscillation is approximately:

$$T = 0.686\,\mathrm{s}$$

Derivation of the Differential Equation for Damped Harmonic Motion

Consider a block of mass m attached to a spring with spring constant k that oscillates while experiencing a damping force F_d. The damping force is assumed to be proportional to velocity:

$$F_d = -bv = -b\frac{dx}{dt}$$

where b is the damping constant (with SI units kg/s). The restoring force due to the spring is given by Hooke's law:

$$F_s = -kx.$$

Applying Newton's second law along the direction of motion:

$$m\frac{d^2x}{dt^2} = -b\frac{dx}{dt} - kx.$$

Rearranging,

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0.$$

This is the differential equation governing damped harmonic motion.

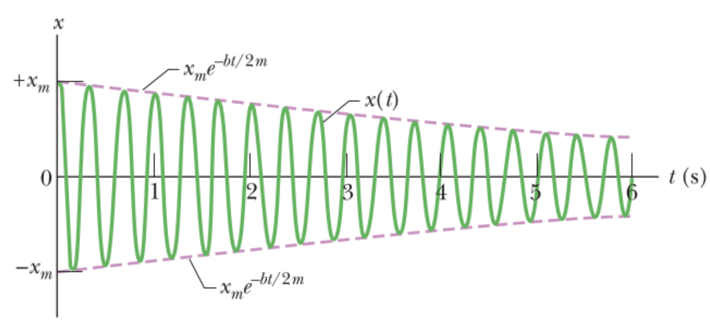


Figure 3: The displacement function x(t) for the damped oscillator.The amplitude, which is xm e!bt/2m, decreases exponentially with time.

Key Properties of Damped Oscillations

1. Displacement Function

The solution to the differential equation is:

$$x(t) = x_m e^{-bt/2m}\cos(\omega' t + \phi),$$

where x_m is the initial amplitude and ϕ is the phase constant.

2. Angular Frequency

The angular frequency of the damped oscillator is given by:

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}.$$

If $b = 0$, then the system reduces to simple harmonic motion with angular frequency $\omega = \sqrt{k/m}$.

3. Energy of the System

The total mechanical energy of the damped oscillator decreases exponentially with time:

$$E(t) = \frac{1}{2} k x_m^2 e^{-bt/m}.$$

The factor $e^{-bt/m}$ represents the gradual dissipation of energy due to damping.

Problem 1:

For the damped oscillator , the given parameters are:

$$m = 250 \text{ g} = 0.25 \text{ kg}, \quad k = 85 \text{ N/m}, \quad b = 70 \text{ g/s} = 0.070 \text{ kg/s}.$$

(a) What is the period of the motion?

(b) How long does it take for the amplitude of the damped oscillations to drop to half its initial value?

(c) How long does it take for the mechanical energy to drop to half its initial value?

Solution

(a) Period of the Motion

The angular frequency for a damped oscillator is given by:

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}.$$

Substituting the given values:

$$\omega' = \sqrt{\frac{85}{0.25} - \frac{(0.070)^2}{4(0.25)^2}} = \sqrt{340 - 0.0196} \approx \sqrt{339.98}.$$

Approximating,

$$\omega' \approx 18.44 \text{ rad/s}.$$

The period is given by:

$$T = \frac{2\pi}{\omega'} = \frac{2\pi}{18.44} \approx 0.34 \text{ s}.$$

Answer:

$$T \approx 0.34 \text{ s}.$$

(b) Time for Amplitude to Drop to Half

Key Idea: The amplitude of a damped oscillator follows the exponential decay:

$$x_m(t) = x_m e^{-bt/2m}.$$

To find the time when the amplitude drops to half:

$$\frac{x_m}{2} = x_m e^{-bt/2m}.$$

Canceling x_m:

$$\frac{1}{2} = e^{-bt/2m}.$$

Taking the natural logarithm on both sides:

$$\ln \frac{1}{2} = -\frac{bt}{2m}.$$

Solving for t:

$$t = -\frac{2m}{b} \ln \frac{1}{2}.$$

Substituting values:

$$t = -\frac{2(0.25)}{0.070} \ln \frac{1}{2}.$$

Since $\ln \frac{1}{2} = -\ln 2$, we get:

$$t = \frac{2(0.25)\ln 2}{0.070} \approx \frac{0.5 \times 0.693}{0.070}.$$

$$t \approx \frac{0.3465}{0.070} \approx 5.0 \text{ s}.$$

Answer:

$$t \approx 5.0 \text{ s}.$$

(c) Time for Mechanical Energy to Drop to Half

$$E(t) = \frac{1}{2} k x_m^2 e^{-bt/m}.$$

To find the time when energy is halved:

$$\frac{E_0}{2} = E_0 e^{-bt/m}.$$

Canceling E_0:

$$\frac{1}{2} = e^{-bt/m}.$$

Taking the natural logarithm:

$$\ln \frac{1}{2} = -\frac{bt}{m}.$$

Solving for t:

$$t = -\frac{m}{b} \ln \frac{1}{2}.$$

Since $\ln \frac{1}{2} = -\ln 2$, we get:

$$t = \frac{m \ln 2}{b}.$$

Substituting values:

$$t = \frac{0.25 \times 0.693}{0.070}.$$

$$t \approx \frac{0.173}{0.070} \approx 2.5 \text{ s}.$$

Answer:

$$t \approx 2.5 \text{ s}.$$